Environmental Brain Cube (EBC)

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Personal Project Report

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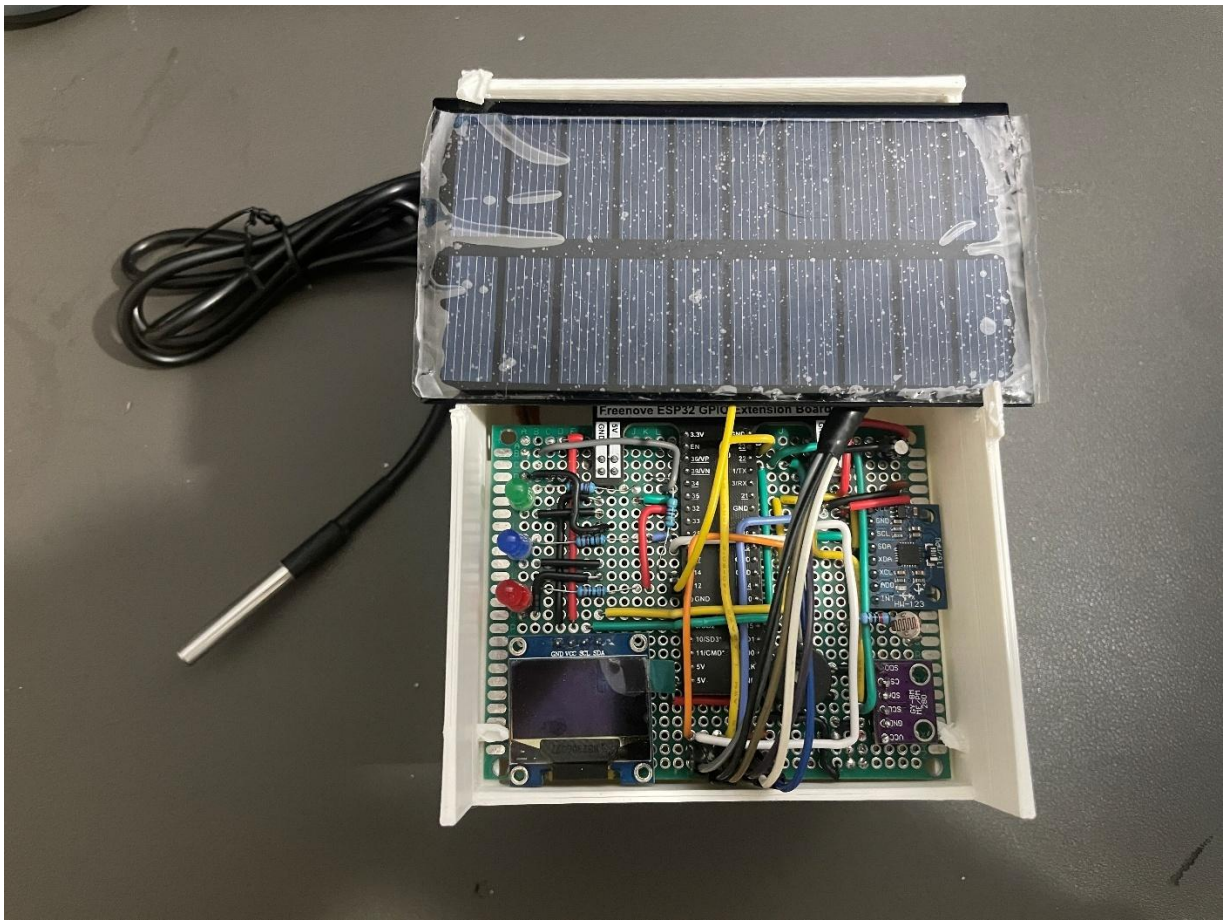

Abstract

The Environmental Brain Cube (EBC) is a compact environmental monitoring and evaluation system. It is designed to analyze environmental conditions and make conclusions per user defined parameters. The system makes use of multiple sensors to measure air temperature, ground temperature, pressure, humidity, motion, and light intensity. All these measurements are displayed and logged to an SD card in real time.

The primary goal of this project is to see whether a system can autonomously make correct conclusions of a given environment for long term use. The EBC was tested in simulated environments of normal, warm-humid, warm-dry, and cold. These conditions were set in controlled container setup. Output data was then analyzed using MATLAB to identify trends and predict the environment's long-term behavior.

Results showed that the system can correctly classify environments and determine suitability of the system in a set environment. This demonstrates the foundation for applications in space exploration as well as environmental monitoring systems.

1. Introduction

In a lot of engineering applications, understanding and adapting to environmental conditions is very important for reliable operation. A system set in an unknown environment is vulnerable to failure due to harsh conditions it was not prepared for. This is especially important in space exploration, where factors such as temperature, pressure, and humidity can significantly impact system functions and reliability. It is therefore important to collect data about a given environment to better understand it and determine if any systems meant to run in this environment will safely do so.

The purpose of this project is to develop an integrated system capable of sensing, monitoring, and evaluating given environmental conditions in real time. The Environmental Brain Cube allows for such capabilities, by using a multi-sensor platform to collect data, identify trends, and determine whether a given environment is suitable for long term operation. This system provides a comprehensive understanding of both ambient and surface level conditions.

To validate the system's results, controlled experiments were conducted using a simulation box with different environmental conditions including normal, warm-humid, warm-dry, and cold. The data collected during these tests is logged and analyzed to evaluate system performance and determine how accurately the EBC can classify an environment and predict suitability for long term use.

2. System Overview

2.1 Hardware components

The system consists of sensing, processing, power, storage, and output components integrated into one embedded system:

Table 1: EBC Component list

Component	Description
ESP32 WROVER (Microcontroller)	Dual-core processor used for data acquisition, control logic, power management, and communication between all components; operates at 3.3V
BME280	Measures ambient temperature (°C), relative humidity (%), and atmospheric pressure (hPa)
MPU6050	Measures motion responses including acceleration (x, y, z axes) and angular velocity (gyroscope)
DS18B20	Measures ground temperature (°C) on which the system operates
5V Solar Panel	Reads voltage variations due to lighting conditions to estimate theoretical power output
9V Lithium Battery	Serves as the primary power storage source for the system
LEDs	Provide visual indicators when sensor readings exceed defined thresholds
Active Buzzer	Produces an audible alert when any system parameter crosses set limits
Photoresistor	Measures light intensity to classify environmental lighting conditions
OLED Display (SSD1306)	Displays real-time sensor data and system status
SD Card Module (128GB SD Card)	Logs collected data for storage and further analysis

Sensor specifications and operating characteristics were obtained from manufacturer datasheets [1]–[3].

2.2 Voltage Divider

To safely measure voltages above the ESP32's 3.3V ADC limit, voltage dividers were implemented for both the battery and solar panel.

Each divider consisted of two resistors:

- Top resistor ≈ 100 kΩ
- Bottom resistor: ≈ 47 kΩ

The output voltage to the ESP32 is given by:

$$V_{out} = V_{in} \times \frac{R_{bottom}}{R_{top} + R_{bottom}}$$

This relation is based on standard voltage divider principles [5]

This ensures that higher input voltages, like those of the 9V battery and 5V solar panel, are scaled down to a safe measurable range. The ESP32 reads this reduced voltage, and the original voltage is reconstructed using the inverse relation.

2.3 Physical Construction and Integration

The system was assembled using a combination of soldered wiring on perfboards and a custom housing design.

Circuit Assembly:

- Components were mounted on perfboards for structural stability
- Electrical connections were made using soldered wires
- Power and ground lines were distributed across common rails

Resistors and Supporting Components:

- 4.7 kΩ resistor used as pull-up for the DS18B20 data pin
- 220 Ω resistors used for current limiting on LEDs
- Voltage divider resistors (~100 kΩ and ~47 kΩ) used for voltage sensing
- ~10 kΩ resistor used in the photoresistor voltage divider

Capacitors

- 10 µF capacitors placed near power inputs (ESP32 and SD card module and Solar panel) to stabilize voltage
- 0.1 µF capacitors used for noise filtering in sensitive analog pins

Housing Design

The enclosure was designed using Autodesk Inventor and fabricated using FDM 3D printing on an Ender 3D printer. It was printed using PLA material to provide structural support and protection for all components. Its main purpose was to organize internal wiring to reduce interference and to allow access to sensors while maintaining compactness.

The final assembled system integrated all electrical and mechanical components into a compact, portable unit suitable for controlled environmental testing.

2.4 Architecture

The system follows a layered process of sensing, processing, decision making, and outputting:

Data Acquisition

- Sensors collect environmental data:

    - BME280 – temperature, humidity, and pressure
    - DS18B20 – ground/surface temperature
    - MPU6050 – motion and vibration
    - Photoresistor – light intensity
    - Voltage dividers – monitor power system readings from the solar panel and battery

Processing

- Raw data is collected by the ESP32 from the sensors and converted to engineering units
- Any invalid or missing data is filtered to provide a reasonable reading
- Data is then organized for further calculations and logging

Evaluation

- The system evaluates the processed data against given thresholds
- For environmental classification, processed data is compared with set environment types, and the best match is used as the final decision
- For environmental suitability, processed data is compared with set thresholds to see if it stayed within those thresholds and how much the offsets were.

Output

- OLED Display:
    - Shows a real-time visualization of measurements being taken or processing being conducted
- Serial Monitor:

    - Like the OLED display, the serial monitor shows a more in-depth real time visualization of measurements and processes
- SD Card Logging:
    - All collected data including result summaries and conclusions are logged to the SD card as csv and text files.
- Warnings:
    - Any values that cross past a threshold are flagged and activate a visual (led ON) and audible (buzzer sound) output

Post-Processing

- Logged csv data is imported to MATLAB where it will be further processed to generate graphs and comparison tables which will be used for trend analysis and long-term predictions

3. Experimental Setup

3.1 Simulation Box

Experiments were conducted using a transparent plastic container with a lid, that served as a controlled environment enclosure. This allowed conditions to be stabilized for the duration of a test. The Environmental Brain Cube was first placed in a small open container with sand where the DS18B20 was to be placed for ground temperature readings, then placed inside the main test container.

3.2 Environmental Conditions

Different environments were made using these elements:

- **Normal**: No additions (left to operate alone in the box)
- **Warm-Dry**: A warm blower blew warm air into the container through a small opening. This allowed for dry air to enter and reduces humidity (~15–35%).
- **Warm-Humid**: A bowl with hot water was set inside the container to increase temperature and humidity (~60–70%)
- **Cold:** The simulation box was placed in a freezer to cool down(~4–12°C)
- **Light:** A desk lamp was used to simulate light input

All conditions were allowed to stabilize before data was recorded.

4. Methods

4.1 Test Procedure

Each test followed the same procedure

1. Place the EBC system inside the simulation box with the selected environment set up
2. Start the test run and begin data display and logging
3. Record the measured data to the SD card throughout the run
4. Monitor the system for any warnings or faults
5. End the test after the set duration and view the results

4.2 Data Collection

Data was continuously collected and stored on a time basis. This was in the form of a csv file uploaded to the SD card for later analysis. Minimum and maximum values for each parameter were collected for future average computation.

Logged parameters:

- Air temperature (BME280)
- Ground temperature (DS18B20)
- Humidity
- Pressure
- Light level
- Acceleration (motion and vibration)
- Battery voltage
- Solar panel voltage

5. Software and Logic

5.1 Arduino code system logic

The ESP32 system runs a continuous loop to monitor environmental and system parameters. The control code was developed using the Arduino IDE [4].

- Sensor readings are taken for air temperature, ground temperature, humidity, pressure, light level, acceleration, and voltage

- Each value is compared against predefined threshold limits
- Warnings are triggered if limits are exceeded
- All data is logged to the SD card with time tracking for later analysis
- The system uses non-blocking timing to ensure consistent and reliable data collection

5.2 Mode/Comparison Logic

The system compares collected data against predefined environmental profiles to match the best profile with the test conditions that were set.

Table 2: Environment Profiles and their respective parameters

Mode	Temperature(°C)	Humidity (%)	Pressure (hPa)
Normal	20.0 – 24.0	30.0 – 50.0	995.0 – 1025.0
Warm-Dry	24.0 – 30.0	15.0 – 35.0	995.0 – 1025.0
Warm-Humid	26.0 – 32.0	60.0 – 70.0	995.0 – 1025.0
Cold	4.0 – 12.0	35.0 – 65.0	995.0 – 1025.0

At the end of a run, average values are calculated and compared to the ranges above for each profile. The system determines which environment closely matches and a rating is assigned based on how many parameters fall within the expected range.

5.3 Suitability Logic

The suitability logic determines whether the environment is safe for the system to run in for the duration of the test run, as well as for future long-term use. The user sets environment parameters for their needs and the system considers environment safe if all parameters of the test

environment meet those of the set environment. Failure occurs when a parameter exceeds limits for a continuous set time (not just a short spike). This triggers the system to deem the environment unsafe and stop the program.

5.4 Prediction Logic

Prediction logic estimates long term behavior of an environment based on trends observed during a run. Stable values that stabilize over time indicate that the environment is likely safe and steady for continued use. Short spikes or disturbances are seen as external effects and are not used in prediction calculations. However, continuous rising or falling trends are considered warnings of instability for long term use of the system. Based on observed behavior, the system determines if parameters will either stabilize around the most recent steady values or continue trending toward the system limits. If a parameter shows continued increase or decrease, the system estimates whether it may eventually reach set limits in the future.

6. Data Analysis

6.1 Graphs

MATLAB was used to create visualizations of the recorded parameters [6], which helped to illustrate the system's behavior during each test. These parameters included air temperature, ground temperature, humidity, pressure, light level, acceleration, battery voltage, and solar voltage. Each graph showed how the values changed over the duration of the test. For example, a

temperature plot for the warm-dry condition is shown in Figure 1 (Appendix), while humidity and pressure plots are provided in Figures 2 and 3 (Appendix).

6.2 Environmental Matching

The collected data helped classify the test environment by comparing what was measured against a set of predefined environmental conditions. To do this, average temperature, humidity, and pressure values were calculated and then checked against the expected ranges for each mode. Whichever environment came closest to the measured data was chosen as the match.

The plotted data in the Appendix backs up this comparison, allowing the relationship between the measured values and the expected ranges to be visualized. The final classification shows how closely the system's recorded conditions lined up with what the experimental setup was designed to produce. A summary of the test run was also produced for each matching evaluation, as seen in Figure 4.

6.3 Trend Analysis

Trend analysis was done by looking at the shape and progression of the plotted data over the duration of the run. Parameters that appeared to stay rather constant or settled down were taken as a sign of stability, suggesting that the environmental conditions were consistent throughout the run. On the other hand, values that drifted gradually up or down were looked at more closely to figure out whether they were genuine environmental changes or were simply brief disturbances.

Patterns that stood out, like a value stabilizing after an initial jump, or a steady drift in one direction, were used to get a better read on how the system was behaving. Such trends can be seen in the plotted results included in the Appendix, with Figure 5 being a good example.

6.3 Predictions

Predictions about future system behavior were drawn from the trends observed in the MATLAB plots. For parameters that showed stable behavior, it was assumed they would continue to sit within a steady range going forward. Where a continuous rise or fall was spotted, the system estimated whether the parameter would eventually approach or push past defined limits, and when relevant, a rough time frame for reaching those limits was worked out based on how quickly the value was changing. For parameters that showed no real long-term trend, no prediction of limit exceedance was needed.

Examples of this prediction behavior, along with trend extensions, are shown in the Appendix, with Figure 5 being a good illustration where the prediction is overlaid directly onto the measured data. A suitability summary at the end of suitability test run evaluations showed an overview of performance and predictions, as shown in Figure 6.

7. Results and Discussion

The results from the experimental runs show that the Environmental Brain Cube (EBC) was able to successfully monitor and record environmental conditions across all tested scenarios. The collected data and the MATLAB analysis gave a clear picture of the system's performance, environmental behavior, and overall suitability.

For the different environments, the system operated steadily, with all most parameters staying within defined safe thresholds. As expected, the warm and warm-humid environments produced noticeable increases in both air and ground temperature, with humidity levels rising significantly

in the warm-humid case. These trends are visible in the temperature and humidity plots shown in Figure 9 and 10 in the Appendix. The warm-dry environment, on the other hand, showed high temperatures but reduced humidity, confirming that the environmental controls used in the simulation box were effective in making a warm-dry environment and the system correctly identified that environment.

The cold environment led to lower temperature and humidity readings. All the parameters remained within acceptable ranges throughout the test. No critical failures were observed in runs where conditions stayed within threshold limits, suggesting that the system can hold up across a range of moderate environments.

Trend analysis revealed that many parameters either stayed stable or showed minor fluctuations before settling down. This shows that some disturbances were caused by external factors rather than any underlying environmental instability. In cases where parameters showed a continuous rise or fall, the prediction logic flagged this as unsuitable for long term use if those trends were to continue. These behaviors can be seen in the plotted data with prediction overlays, such as he voltage plot in Figure 7 in the Appendix, which shows that for long term use, the battery voltage would decrease significantly and won't power the system.

The environment matching logic further confirmed that the measured data lined up closely with the intended test conditions. In most cases, the system correctly classified the environment based on average temperature and humidity values, showing that both the experimental setup and the comparison logic were effective in producing and identifying distinct environmental profiles.

Overall, the results indicate that the EBC system is well capable of reliably collecting environmental data, identifying trends, and evaluating suitability. While the simulation setup

offers a simplified model of real world conditions, the behavior observed throughout testing suggests that the system could be extended to more complex environments with further adjustments.

8. Conclusion

The Environmental Brain Cube (EBC) successfully demonstrated the ability to monitor, record, and analyze environmental conditions within a controlled simulation setup. Across all test environments, the system reliably collected data and provided meaningful insights into temperature, humidity, pressure, and parameters such as power behavior.

The results showed that the system performs well under stable conditions and can distinguish between different environmental profiles. The implemented suitability and prediction logic allowed the system to assess whether an environment is safe for continued operation and identify potential long-term risks based on observed trends.

Overall, the project validates the concept of a compact environmental monitoring system capable of supporting decision making in different environments, for use in applications such as space exploration, where systems must operate reliably in extreme and unpredictable environments

9. Limitations

While the system performed effectively, several limitations were identified:

- The simulation box represents a simplified environment and does not fully replicate real world conditions
- Environmental changes were manually introduced, which may lead to inconsistencies between runs
- Sensor accuracy and resolution (e.g. DHT11 limitations) may affect precision of collected data
- Power calculations are based on voltage measurements and estimated assumptions rather than direct current sensing
- Prediction models are trend based and do not account for complex physical interactions or sudden environmental changes

These limitations should be considered when interpreting results and extending the system to real applications.

10. Improvements

Several improvements can be made to enhance the system's performance and applicability:

- Integrate current sensing for more accurate power and energy calculations
- Improve environmental control using automated heating, cooling, and humidity systems

- Use higher precision sensors to improve data accuracy
- Expand prediction models to include more advanced methods (e.g. machine learning)
- Test the system in more realistic outdoor or field environments

Future developments will focus on increasing realism, accuracy, and system autonomy.

Appendix

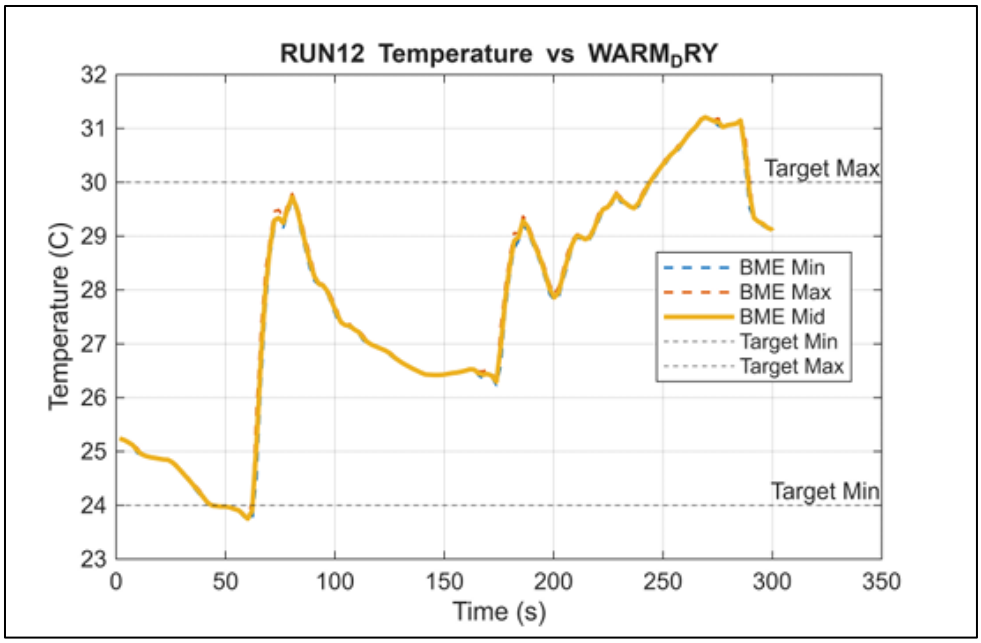

Figure 1: Plot of run 12 temperatures against warm-dry threshold

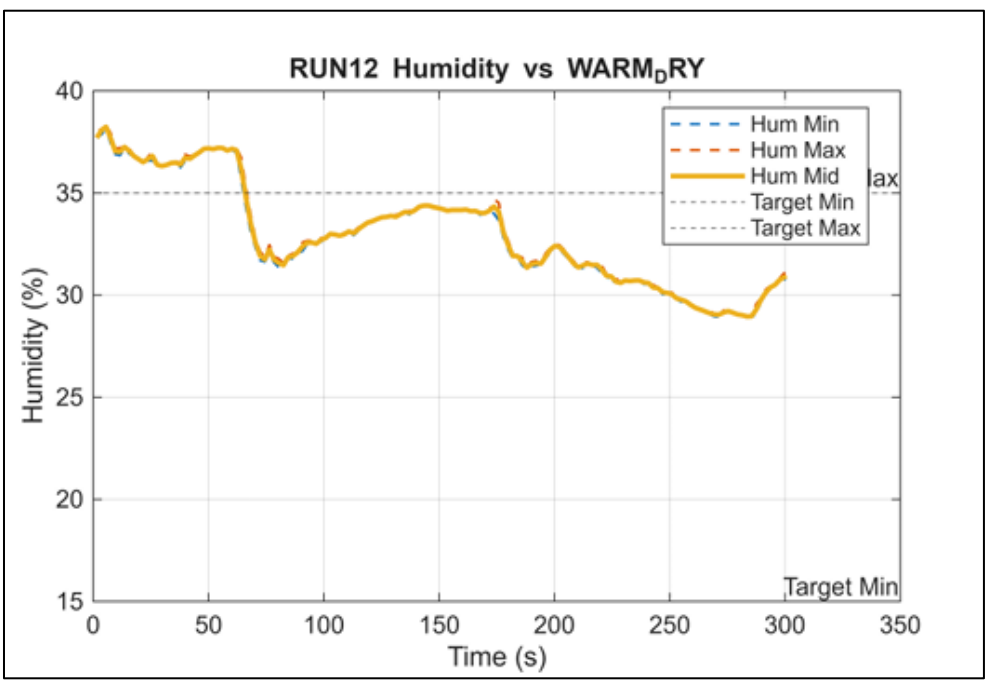

Figure 2: Plot of run 12 humidity against warm-dry threshold

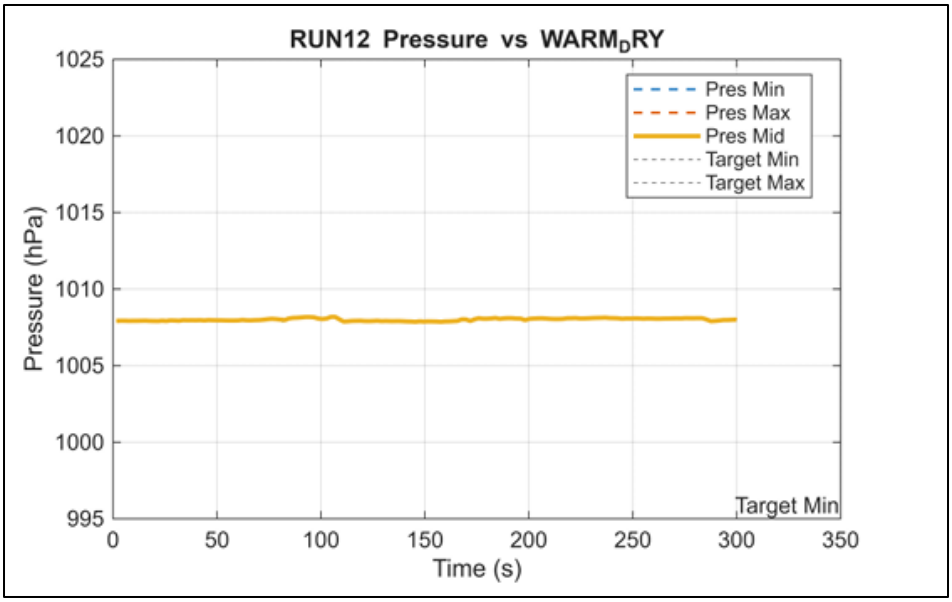

Figure 3: Figure 2: Plot of run 12 pressures against warm-dry threshold

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================ CONDITION COMPARISON ================
Run analyzed: RUN12
Best matched condition: WARM_DRY
Run duration: 300.01 s
Average run values used for comparison:
- BME temperature average: 27.72 C
- DS temperature average: 20.06 C
- Humidity average: 32.97 %
- Pressure average: 1008.01 hPa
Best matched condition ranges:
- Temperature: 24.0 to 30.0 C
- Humidity: 15.0 to 35.0 %
- Pressure: 995.0 to 1025.0 hPa
Parameter results:
- Temperature: PASS | closeness 76.1 %
- Humidity: PASS | closeness 20.3 %
- Pressure: PASS | closeness 86.7 %
Overall:
- Match count: 3/3
- Overall closeness: 61.0 %
- The run fits the matched condition well overall.
Environment ranking from best to worst:
1. WARM_DRY | match 3/3 | closeness 61.0 %
2. WARM_HUMID | match 2/3 | closeness 48.0 %
3. NORMAL | match 2/3 | closeness 38.8 %
4. COLD | match 1/3 | closeness 28.9 %
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Figure 4: Condition comparison summary at the end of environment classification for run 12

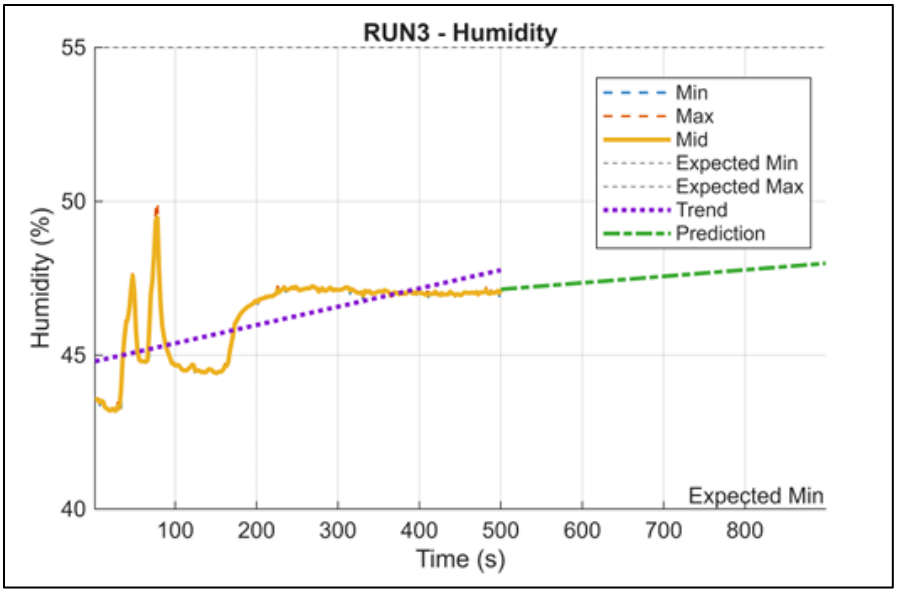

Figure 5: Suitability test for run 3 showing humidity examination

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============= SUITABILITY ANALYSIS =============
Run: RUN3 | Duration: 500.01 s | Format: new_min_max
Averages:
- Air: 22.78 C | Ground: 20.45 C | Humidity: 46.28 % | Pressure: 1008.13 hPa
- Battery: 5.72 V | Solar: 4.79 V | Light: 3781.91
Core check:
- Air 88.7% | Ground 17.9% | Humidity 83.8% | Pressure 83.7%
Power check:
- Battery avg power: 0.515 W
- Solar avg power est: 0.957 W
- Net margin: 0.443 W
- Battery power vs drop: ALIGNS
- Solar power vs trend: HIGH BUT DROPPING
- Est. battery charge time from solar: 6.58 hr ideal, 8.23 hr adjusted
Trend notes:
- Air: drift = -0.810, directional consistency = 0.62, behavior = STABLE | no boundary estimate needed
- Ground: drift = -0.324, directional consistency = 0.20, behavior = STABLE | no boundary estimate needed
- Humidity: drift = 1.892, directional consistency = 0.53, behavior = STABLE | no boundary estimate needed
- Pressure: drift = 0.111, directional consistency = 0.44, behavior = STABLE | no boundary estimate needed
- Battery: drift = -0.169, directional consistency = 0.36, behavior = STABLE | no boundary estimate needed
- Solar: drift = -0.154, directional consistency = 0.41, behavior = STABLE | no boundary estimate needed
Other notes:
- Avg air-ground temp diff: 2.335 C
- Air-ground temp correlation: 0.953
Result:
- Long-term score: 68.5 %
- Decision: MARGINALLY SUITABLE LONG-TERM
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Figure 6: Suitability analysis for run 3

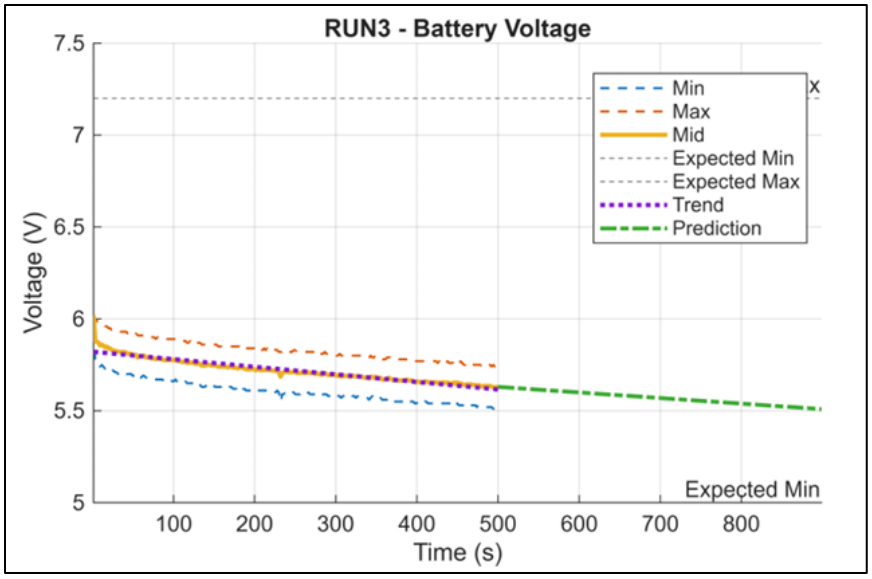

Figure 7: Suitability run 3 data for battery voltage

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================ SIMPLE CONCLUSIONS ================
Run analyzed: RUN8
Total run time analyzed: 2799.76 s
Temperature:
- Average BME temperature midpoint: 22.91 C
- Average DS temperature midpoint: 20.76 C
- Highest temperature observed: 23.91 C
Humidity:
- Average humidity midpoint: 43.95 %
- Highest humidity observed: 56.93 %
Pressure:
- Average pressure midpoint: 1007.78 hPa
Power:
- Average battery midpoint: 6.58 V
- Average solar midpoint: 2.17 V
Warnings:
- No temperature warning intervals occurred.
- No humidity warning intervals occurred.
Light:
- Most common light state was NORMAL.
Motion:
- Motion appears strong or highly variable.
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Figure 8: Free run summary for run 8

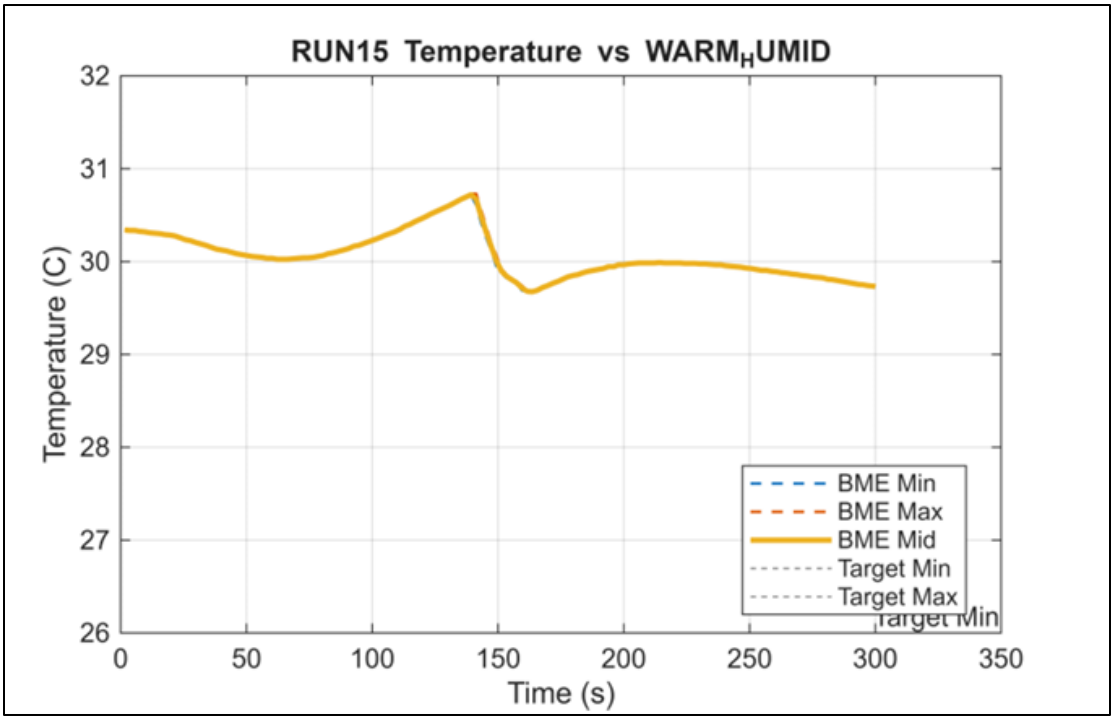

Figure 9: Temperature data for run 15 against set warm-humid conditions

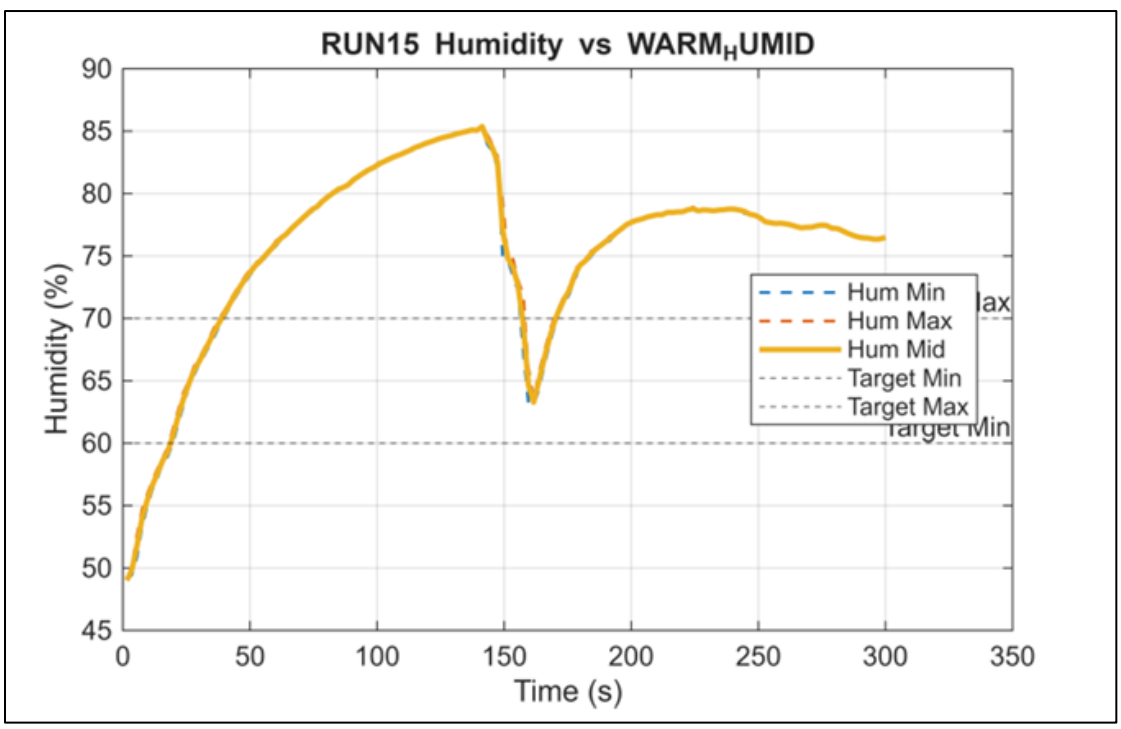

Figure 10: Humidity data for run 15 against set warm-humid conditions

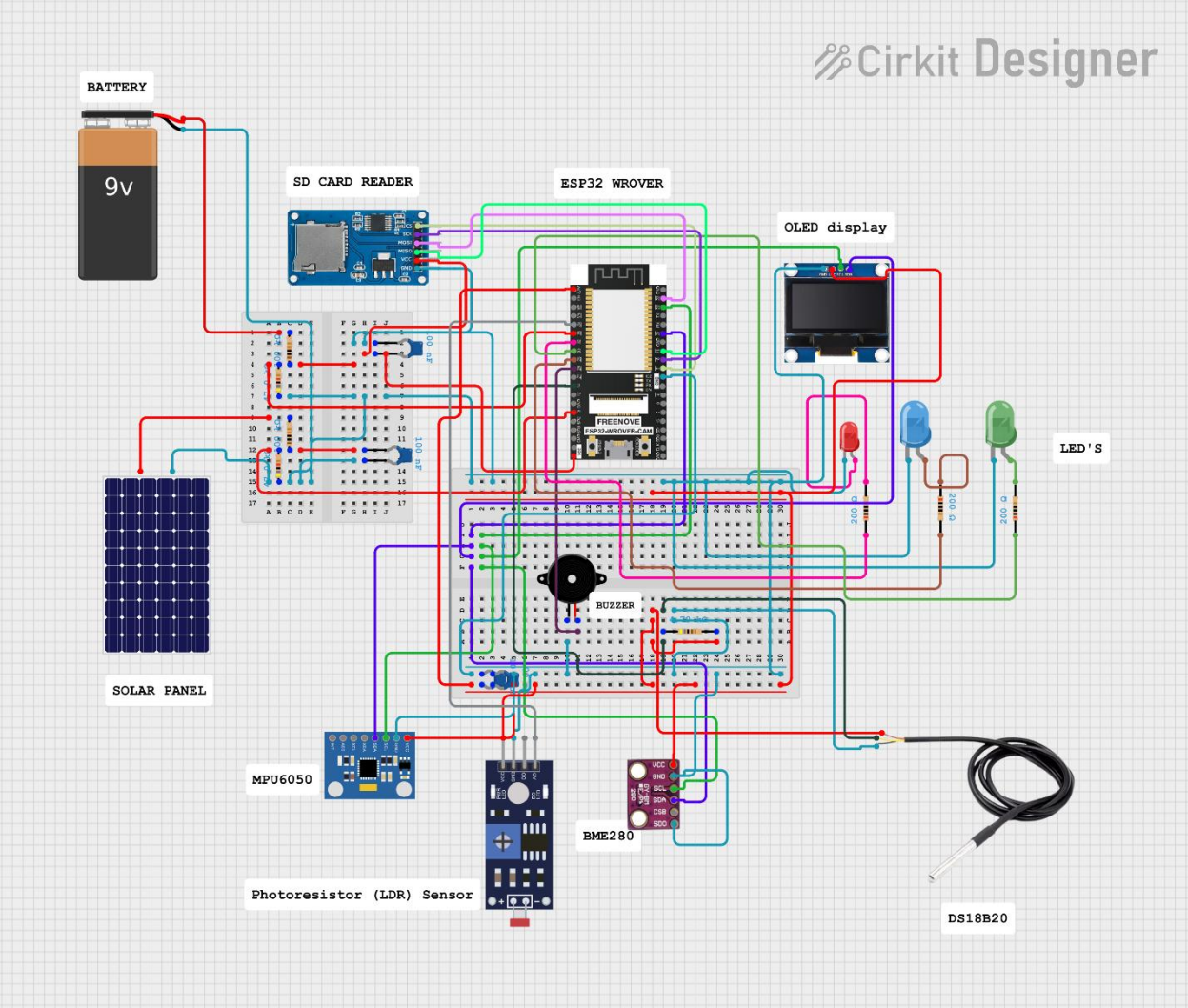

Figure 11: Rough layout of EBC components

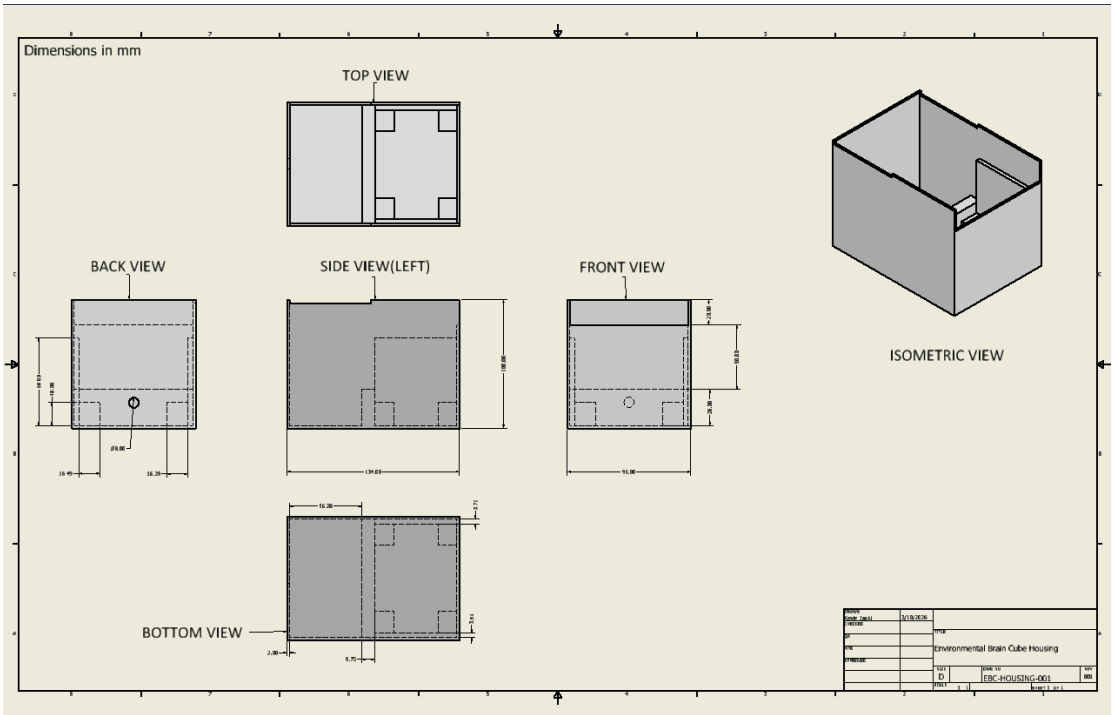

Figure 12: Drawing of 3D-printed housing made from PLA, weighing 44 g, with overall dimensions of 100 mm (height) × 134 mm (length) × 96 mm (width)

Figure 13: Simulation Box

Supplementary Material

All supporting materials including source code, data files, CAD models, and experimental media are provided separately: https://github.com/ktjauki/Environmental-Brain-Cube.git

References

[1] Bosch Sensortec, BME280 Datasheet, 2018.

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[4] Arduino, Arduino IDE Documentation.

[5] J. W. Nilsson and S. A. Riedel, Electric Circuits.

[6] MathWorks, MATLAB Documentation.